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Sleep in healthy elderly and amnestic Mild Cognitively Impaired (aMCI) patients due to neurodegeneration

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Abstract

Both healthy aging and neurodegeneration are associated with sleep impairments. In Alzheimer's Disease (AD), a wide array of sleep abnormalities is present including deficits in both Rapid Eye Movement (REM) and non-REM (NREM) sleep stages. However, it is still unclear if and how sleep is disrupted in early stages of disease, beyond what is expected by age. Amnestic Mild Cognitive Impairment (aMCI) is a syndrome defined by a cognitive decline, most pronounced for memory. aMCI often (~60-65%) progresses to dementia, and is widely considered as a prodromal phase of AD. Only few studies compared sleep in aMCI versus age-matched healthy controls. Here, we collected overnight sleep data from aMCI patients (n=15) and AD dementia patients (n=11) who were clinically diagnosed and referred by the Neurology Memory clinic at the Tel Aviv Sourasky Medical Center (TASMC, 'Ichilov'). Additional data were collected from age-matched healthy participants (n=15) recruited from the community. We performed overnight sleep polysomnography with high-density (256 channel) electroencephalography (EEG), electromyography (EMG), electrooculography (EOG), video, breathing monitoring, as well as cognitive testing including questionnaires and an overnight memory consolidation paradigm with testing before and after sleep. Exclusion criteria included history of neuropsychiatric disorders, use of CNS-acting drugs or breathing abnormalities (moderate OSA excluded). We found that cognition and memory, including clinical questionnaires and overnight memory score at our lab, were significantly degraded in aMCI patients, and more so in AD patients, compared to healthy individuals. The groups did not differ significantly in aspects such as age, gender, or breathing profiles. With respect to sleep, AD patients showed a variety of sleep impairments including longer sleep onset, decreased sleep efficiency and total sleep time, shorter REM and deep NREM sleep stages, which significantly differentiated AD from aMCI. When focusing on differences between aMCI and healthy individuals, a significant difference was observed exclusively in REM sleep stage, manifested as longer REM sleep onset latency. Moreover, overnight memory scores were correlated with REM sleep integrity, such that decreased REM sleep could predict lower overnight memory consolidation. These results suggest that REM sleep represents an early marker of neurodegeneration in aMCI, and points to REM sleep as a specific marker that is sensitive for early detection of neurodegeneration and dementia.

תקציר

הזדקנות בריאה ונורמטיבית וגם נוירודגנרציה מלוות בבעיות וקשיי שינה. במחלת האלצהיימר קיימת קשת רחבה של דפוסי שינה אבנורמליים בכלל שלבי השינה, ביניהם בשלבי השינה REM וNREM. אולם, עדיין לא ברור האם וכיצד שינה מופרעת בשלבים מוקדמים של המחלה, מעבר למה שמצופה מתהליכי ההזדקנות הטבעית. הפרעה קוגנטיבית אמנסטית מתונה (aMCI) היא סינדרום המאופיין בהתדרדרות קוגנטיבית המודגשת במיוחד בתחום הזכרון שנחלש. לאינדיבידואלים עם aMCI קיים סיכון מוחשי (~ 60-65%) לפתח דמנציה במהלך השנים שלאחר האבחנה, ולכן קיימת הסכמה רחבה כי הסינדרום מהווה שלב פרודרומלי של מחלת האלצהיימר (AD). רק מחקרים בודדים השוו שינה בחולי aMCI למול קבוצת ביקורת בריאה תואמת גיל. במחקר זה, אספנו דאטה ממהלך שינת לילה שלם מחולי aMCI (n=15) וחולי דמנציית אלצהיימר (n=11) שאובחנו באופן קליני והופנו אלינו על ידי מרפאת הזכרון במחלקת הנוירולוגיה של המרכז הרפואי סוראסקי תל אביב (TASMC, איכילוב). דאטה נוסף נאסף ממשתתפים בריאים תואמי גיל (n=15) שגויסו מהאוכלוסייה. ביצענו פוליסומנוגרפיה לאורך לילה שלם על ידי רשת בעלת 256 ערוצים הכוללת אלקטרואנצפלוגרפיה (EEG), אלקטרומיוגרפיה (EMG) ואלקטרואוקולוגרפיה (EOG), וידאו, וניטור נשימה, כמו גם מבחנים קוגנטיביים הכוללים שאלונים נוירולוגיים מקובלים וכן פרדיגמה הבוחנת התגבשות זכרון לאורך הלילה עם מבחנים לפני ואחרי שינת הלילה. קריטריוני הוצאה אל מחוץ לניסוי כללו היסטוריה של מחלות נוירולוגיות או פסיכיאטריות, שימוש בתרופות הפועלות על מערכת העצבים המרכזית או הפרעות נשימה (הפרעת נשימה חסימתית OSA). מצאנו כי קוגניציה וזכרון, שנבחנו על ידי שאלונים קוגנטיביים וציון של התגבשות זכרון לאורך הלילה במעבדת השינה שלנו, היו נמוכים משמעותית בחולי aMCI, ואפילו יותר בחולי אלצהיימר, בהשוואה לקבוצת הבריאים. הקבוצות לא הראו הבדלים ביניהן בפרמטרים כגון גיל, מגדר או פרופיל נשימה. בתחום השינה, חולי אלצהיימר הראו מגוון בעיות שינה הכוללות זמן שינה מופחת, יעילות שינה נמוכה, זמן שינה מופחת בשלבי ה-REM והשלבים העמוקים יותר של שינת ה-NREM, שגם הבדילו באופן מובהק את קבוצת ה-aMCI מחולי האלצהיימר. בהתמקדות על ההבדלים בין חולי aMCI וקבוצת הביקורת הבריאה תואמת הגיל, הבדל משמעותי נצפה רק בשינת ה-REM שהתבטא בזמן ארוך יותר עד לאפיזודת ה-REM הראשונה בלילה. בנוסף, ציוני זכרון לאורך הלילה היו בקורלציה משמעותית עם כמות שינת ה-REM, כך שהפחתה בשינת ה-REM יכולה לנבא ציונים נמוכים יותר במבחן התגבשות הזכרון לאורך הלילה. תוצאות אלו מרמזות כי ייתכן והזמן עד לכניסה לשינת REM מהווה סמן מוקדם וייחודי לנוירודגנרציית אלצהיימר, ומצביעות על שינת ה-REM כמרקר יעודי שהינו רגיש במיוחד לזיהוי מוקדם של נוירודגנרציה ודמנציה.

Introduction

Sleep

Sleep is a state of reduced consciousness and immobility, defined by a reversible disconnection from the environment that is homeostatically regulated. Normal sleep in humans occurs on a daily basis and varies throughout an individual's lifespan, usually consisting of a few 90 minute cycles each including both rapid eye movement (REM) sleep and non-REM (NREM) sleep components, with NREM sleep further divided into stages 1 thorough 3 (N1, N2 & N3), in which sleep gradually deepens (Markov and Goldman, 2006).

As sleep is initiated, EEG's waking frequency gradually slows down and as sleep further deepens high-amplitude (> 75 μV in scalp recordings) slow waves (0.5–4-Hz) emerge. N3 is the deepest stage of sleep and is referred to as slow-wave sleep (SWS) since it is characterized by high prevalence (>20% of time) dominated by slow wave activity. In humans, NREM is characterized by reduced neuronal activity, and dream reports during this state in humans only consist of ruminative nonvisual thoughts (Markov and Goldman, 2006).

The first REM period usually occurs later in sleep (~70 min after sleep onset), and is strongly associated to vivid dreaming and includes characteristic physiological changes, such as high level of brain activity resembling wakefulness, rapid eye movements, suppression of muscle tone, reduced thermal regulation, irregular cardio and breath functions and alteration of autonomic nervous tone (Markov and Goldman, 2006). REM sleep is also associated with dramatic changes in cellular neuronal activity including gene expression and neurotransmitter activity (Tseng et al., 2010); the initiation of REM sleep is mainly governed by cholinergic neurons in the pons (Yu et al., 2009) as well as dopaminergic signaling in the basolateral amygdala (Hasegawa et al., 2022).

Sleep stages differ not only in terms of depth of sleep, but also in terms of frequency and intensity of dreaming, EEG oscillations, eye movements, muscle tone and neuromodulation of cortical circuits and local brain activity (McCarley, 2007). Sleep is also characterized by activation and communication between memory systems, while REM, N2 and SWS have all been implicated in sleep-dependent memory processing, as have unique patterns of synchronous neuronal activity characterizing these stages, such as sleep spindles in N2, slow-wave activity of SWS, ponto-occipitogeniculate waves and theta rhythms in REM sleep (Stickgold, 1998).

Since sleep is conserved in most species studied carefully so far, it is assumed it must serve a vital function (Tononi and Cirelli, 2014). Functions theorized to provide satisfactory explanations as for sleep necessity include the brain restitution hypothesis (Xie et al., 2013), synaptic homeostasis and downscaling importance (Tononi and Cirelli, 2014) and even a proposed role in repairing DNA damage (Zada et al., 2021, 2019). Among those functions is a theory concerning the importance of sleep as a memory enhancer (Diekelmann and Born, 2010; Stickgold, 2005); for learning and preserving neurocognitive functions. Different types of learning are associated with sleep-related memory consolidation mechanisms that occur at different brain regions and stages of sleep (REM or non-REM) throughout the night (Fogel et al., 2007).

Sleep, learning and memory

Memory formation includes information acquisition/encoding and retrieval/recall, as well as consolidation that occurs mainly during rest and sleep, which plays a vital rule in forming long-lasting memories. During sleep, labile memories encoded during wakefulness are stabilized and integrated into the long term memory storage (Diekelmann and Born, 2010) through consolidation, in which neocortical connections are strengthened and hippocampal dependence is reduced.

Declarative memories, that are defined by recollection of episodes and facts, rely on hippocampal-mediated connections that store memory fragments across various neocortical regions (Eichenbaum, 2001). Declarative memory deficits can result from impaired encoding, consolidation, storage, retrieval, or a combination of these memory components.

Declarative memory consolidation is believed to depend on sleep mechanisms such as slow-wave activity and spindle activity in NREM stages (Clemens et al., 2005) and theta power that predominates rapid-eye-movement (REM) sleep (Nishida et al., 2009). Also, fluctuating levels of neuromodulators during non-REM and REM sleep such as acetylcholine (Power, 2004) and noradrenaline (Kjaerby et al., 2022) may mediate hippocampal-neocortical information exchange and synaptic plasticity.

Studies imply a role for NREM sleep in the long-term consolidation of episodic memories (Diekelmann and Born, 2010; Walker, 2009). Relative to wake time, NREM sleep beneficially reduces fading of episodic memory representations over time, resulting in better retention (Mander et al., 2013). Mechanistically, these findings fit the theoretical framework of hippocampal-neocortical memory consolidation (Buzsáki, 1996), working through a possible synchronization between thalamocortical slow wave and spindle oscillations and hippocampal rhythms such as ripples (Muehlroth et al., 2019).

Recent research have also highlighted the association between REM sleep and some specific forms of memory (Fogel et al., 2007). For example, REM sleep considered to be crucial for perceptual learning (Censor et al., 2006), visual-spatial memory and specific task-learning (Stickgold and Walker, 2007).

Given that sleep seems to be linked to learning and cognitive functioning through involvement of sleep microstructure, synaptic plasticity and neurotransmitters levels (Maestri et al., 2015), it is plausible that disrupted memory processing during sleep may contribute to memory problems in neurodegenerative diseases and in healthy aging (Westerberg et al., 2012).

Sleep and aging

Aging processes are accompanied with changes in sleep, usually associated with diminished sleep duration, enhanced sleep fragmentation and awakenings, increased time spent in lighter stages of NREM sleep, and less deeper sleep stages, such as slow wave sleep (SWS; Mander et al., 2017). Side by side, a progressive loss of coordination occurs among circadian rhythms resulting in an advanced phase of the circadian cycle, earlier sleep onset and morning waking in elderly individuals.

Cognitive aging is significantly related to sleep integrity (Walker, 2009). For example, age-related volume reductions of the ventrolateral preoptic (VLPO) nucleus (Lim et al., 2014), an anterior hypothalamic structure vital for the initiation and maintenance of sleep, may lead to the increased sleep fragmentation and slow-wave sleep (SWS) deficits found in older adults, known to correlate with memory impairments (Muehlroth et al., 2019; Power, 2004). Therefore, evidence supports that neural systems governing homeostatic regulation and maintenance of sleep wake cycle might be particularly vulnerable to senescence, so that sleep impairments are caused by this age-related neural damage (Bliwise, 1993; Hanyu et al., 2002; Mander et al., 2017). Yet, it remains unclear whether these associations suggest a causal role for sleep in age-related memory decline.

Sleep and Neurodegeneration

The quality, timing and duration of sleep are essentially important for health, cognitive functioning, and overall well-being. Consequently, reduced sleep quality or disturbed sleep are major contributors to poor health, and associated strongly to chronic disease, mental health and age-related cognitive decline (D'Rozario et al., 2020). In addition to age effects, sleep-wake deficits seems to be pronounced in individuals with dementia (Naismith et al., 2014). This evidence has encouraged interest in the impact of impaired sleep and circadian disruptions on cognitive decrease and risk for dementia in the elderly population.

And yet, the interplay between sleep impairment and neurodegeneration is a puzzling question since although traditionally insomnia and sleep impairments are considered symptoms of neurological and neurodegenerative diseases, sleep disruption frequently precedes the onset of cognitive impairment (Bliwise, 1993).

Studies conducted on animal models have shown that sleep disruptions may result in the accumulation of toxic substrates in brain tissues, thus accelerating neurodegenerative processes (Xie et al., 2013). However, whether sleep problems themselves trigger or exacerbate dementia and neurodegeneration remains a mystery.

Either way, sleep disturbance and circadian disarrangement are the most disturbing behavioral and psychological symptoms of dementia (BPSD) (Yu et al., 2009). Sleep disorders are highly prevalent in age-related neurodegenerative conditions as Parkinson, Dementia with Lewy Bodies, Frontotemporal Dementia, and Alzheimer's disease (AD) (Bliwise, 1993; Musiek and Holtzman, 2016; Pase et al., 2017).

Alzheimer's disease

Alzheimer's disease (AD) is the primary cause of dementia worldwide (Ancoli-Israel et al., 1994), and is a progressive neurodegenerative disorder that results in widespread brain atrophy of both gray and white matter brain regions resulting from neuronal loss (Braak and Braak, 1991). The neuropathological hallmarks of the disease include "positive" lesions; extracellular accumulation of amyloid-β (Aβ) into amyloid plaque and intracellular deposition of hyper-phosphorylated tau into neurofibrillary tangles (Hampel et al., 2004; Kerbler et al., 2015), that cause “negative” lesions such as neuronal and synaptic loss.

According to the amyloid cascade hypothesis, accumulation of Aβ deposits into neuritic senile plaques leads to neuronal death and synaptic malfunctions that damage cognitive function and ultimately may lead to AD (Hardy and Selkoe, 2002). On the other hand, converging evidence traces the AD’s neuropathological trail in the brain as associated with tau pathology (Vogel et al., 2021) and maps tau pathology as the vital factor to AD’s brain damage and cognitive decline (Bejanin et al., 2017; Holth et al., 2019).

AD is defined by an obscure etiology and an evasive onset, a memory deterioration, specifically a progressive hippocampal-dependent explicit memory deficits, variable domains of cognitive impairment and behavioral and psychological symptoms (BPSD) (*Psychopharmacology Bulletin*, 1988); including sleep–wake cycle disturbances and possibly delusion, hallucination, agitation and depression. Many studies attempt to investigate the mechanisms in the origins of BPSD development, including genetic susceptibility and neurotransmitter activity alterations (Tseng et al., 2010).

Cholinergic neurons in the basal forebrain, a gray matter area in the medial and ventral parts of the brain, release the neurotransmitter acetylcholine to numerous different brain regions. Anterior basal forebrain nuclei project to the hippocampus, olfactory bulb, and piriform and entorhinal cortices. Posterior basal forebrain nuclei, including the nucleus basalis of Meynert (Nbm) send projections to the amygdala and the cortex (McGinty and Sterman, 1968; Sterman and Clemente, 1962). Regulation of acetylcholine supply to these brain regions is involved in modulation of plasticity, and cholinergic basal forebrain neurons can influence key behaviors such as attention, learning and memory (Power, 2004).

AD’s early pathological features involve degeneration of these Basal Forebrain neurons, which is suspected to lie in the base of aspects of the neurodegeneration’s cognitive decline (Baskin et al., 1999). Acetylcholine (ACh) levels are considerably decreased in patients with AD and deficiency of ACh is considered to be symptomatic for AD (Yu et al., 2009). The impaired cholinergic innervation strongly contribute to declarative memory impairment (Braak and Braak, 1991), and so this converging knowledge led to the development of the commonly prescribed acetyl cholinesterase inhibitor medication.

Drugs approved for AD therapy are scarce and based mainly on two approaches; N-methyl D-aspartate (NMDA) receptor inhibitor and cholinesterase inhibitors that have become the standard therapy. However, both these medications only aim to slow down symptom progression by improving cognitive function rather than treating the pathology core itself (Baskin et al., 1999; Bissette et al., 1996; Cooke et al., 2006; dos Santos Moraes et al., 2006; Mizuno et al., 2004; Yu et al., 2009).

Sleep in AD

Alzheimer's pathology interferes with sleep physiology. Sleep abnormalities typically observed in AD patients, depending on disease severity, have shown impaired sleep patterns ranging from sleep fragmentation and decrease in sleep efficiency to reductions in SWS, spindle activity (Bliwise, 1993) and/or REM sleep (Hita-Yañez et al., n.d.).

AD’s physiological decline of sleep, particularly NREM sleep duration and quality, is a common feature of senescence, but the onset, severity, and nature of these impairments are all significantly accelerated in those with AD (Mander et al., 2016). Although these sleep disruptions have long been considered robust symptoms of AD, emerging evidence implies that the relationship between AD and sleep disturbance may be causal and bi-directional, representing an integral part of the disease and potentially its treatment (Mander et al., 2016).

Patients with Alzheimer's disease (AD) experience difficulties forming and retrieving memories, and Aβ alone may be insufficient to cause impairments (Villano et al., 2017). Those difficulties may be partially attributed to a disruption in sleep-dependent consolidation, in which normally declarative memory is stabilized across cortical areas (Westerberg et al., 2012). It was recently found that sleep characteristics moderated the relationship between Aβ and memory, with a stronger positive correlation for Aβ and memory disturbance in individuals with sleep impairments. These results imply a protective role of sleep in Alzheimer's disease early stages (Wilckens et al., 2018). Disrupted sleep therefore could represent a mechanistic path through which cortical Aβ associates to hippocampus-dependent cognitive decline in the initial stages of AD progression.

Moreover, sleep disturbances have been hypothesized to cause β-amyloid brain deposits since the early stages of AD (Xie et al., 2013(Musiek and Holtzman, 2016), Aβ concentrations increase during periods of wakefulness and decrease during sleep, resulting in a Aβ diurnal cycle (Holth et al., 2017). Thus, sleep pattern alterations can affect Aβ deposits, suggesting that sleep may play a role in AD pathogenesis (Lucey and Bateman, 2014). The disruption of NREM SWA especially could be integral to AD pathophysiology (Mander et al., 2016), but although NREM sleep associations with β-amyloid are most prominent, of note are emerging links between β-amyloid and REM sleep (Mander et al., 2016).

MCI

Mild cognitive impairment (MCI), a syndrome considered as a pre-dementia phase of AD, is a transitional state between normal aging and dementia, of which it may represent a prodromal stage (Maestri et al., 2015). MCI is a clinical situation of subjective memory complaints and an objective mild cognitive impairment, but without any clear cut evidence of dementia (Westerberg et al., 2012); commonly defined as a period in which the individual shows impairment in one or multiple cognitive domains, but with overall preserved activities of daily living (Guillozet et al., 2003, Yu et al., 2009).

MCI syndrome is heterogeneous in its etiology and clinical presentation, and the main impairment may concern either a single or multiple cognitive domains (Maestri et al., 2015). Two main subtypes of MCI have been identified based on the presence or absence of memory impairments: amnestic MCI (a-MCI) and non-amnestic MCI (na-MCI) (Brayet et al., 2016).

Amnestic MCI exhibit a reduction in the medial and inferior temporal lobes that differs them from control subjects, and from the multiple-domain group that show broader areas of destruction (Whitwell et al., 2007a). The constrained pattern of atrophy in the amnestic MCI groups is in line with the theory that MCI in these subjects could signify prodromal AD. The different patterns in other subgroups imply that these subjects may suffer from a distinct underlying pathology (Whitwell et al., 2007a).

aMCI

Patients with amnestic mild cognitive impairment (aMCI) exhibit confined declarative memory deficits, and many eventually progress to an AD diagnosis (Westerberg et al., 2012), so that aMCI is considered the preclinical or very early stage of AD. The validity of aMCI as a symptomatic pre-dementia phase of AD is supported by conversion rates to AD of 30%- 60% over a 3 years period, and up to 80% at 6 years follow-up (Morris et al., 2001), showing a 4-fold increased risk for the development of AD compared with healthy elderly people (Ganguli et al. 2004), 2-fold compared to na-MCI (Brayet et al., 2016), and so the progression from aMCI to dementia, mostly AD, is considered to be time-dependent.

The belief that aMCI represents the prodromal stage of AD has received strong support from neuropathological (Guillozet et al., 2003), biochemical (Hampel et al., 2004); including tau and amyloid alterations in CSF, neuroimaging studies (Rossini et al., 2006); presenting an increase of low frequency brain activity in MCI, mostly in temporal lobe, and neurophysiological evidence include elevated amyloid plaques, beside reduced hippocampus and basal forebrain volume (Jack et al., 2008).

Studies concluded that medial or inferior temporal lobe measurements are the most accurate ROI in terms of providing the best discrimination between subjects with AD and elders, and so have assumed that these are likely to be the first regions of loss in MCI (Convit et al., 2000). In accordance with this, evidence from neuropathologic and neuroimaging studies demonstrated that AD neurodegeneration already occurs in aMCI patients (Chételat et al., 2005; Guillozet et al., 2003), namely, defined local cortex thinning, and NFT accumulation associated with memory impairment, all mostly pronounced in the medial temporal lobe. And yet, many recent studies have aimed to discover biomarkers in order to develop potential therapeutic management, or determinate the likelihood or timeline of cognitive and functional decline to dementia and its specific type (Maestri et al., 2015), without major success so far.

Sleep and aMCI

The extent to which objective neurophysiological sleep parameters are altered in aMCI patients is still unclear. Several sleep studies focused on NREM sleep markers to try and separate MCI from healthy controls. This focus is logical, particularly in the amnestic subgroup, given that NREM is associated with hippocampal-dependent memory consolidation. However, the extent to which NREM sleep abnormalities are present in aMCI is still under debate. Some studies indicate that SWS is more useful in distinguishing MCI than REM sleep, which is comparable to findings in healthy aging

(Westerberg et al., 2012). Other studies investigated aMCI population and reached conclusions regarding an impairment in NREM sleep stages; N2 or N3 decreased amounts (Brayet et al., 2016; Westerberg et al., 2012). By contrast, other studies concluded that contrary to expectations, MCI patients did not exhibit reduced amount of time in deeper stages of NREM sleep, as opposed to AD patients (Walker at 2009; Gorgoni et al., at 2016).

Yet, evidence implies that existence of sleep disturbances in MCI is elevated beyond normal (D'Rozario et al., 2020), for example a reduced amount of REM sleep in MCI patients was found (Hita-Yañez et al., 2013; Maestri et al., 2015). These results did not correlate to any cognitive or behavioral parameters, and could not exclude the presence of sleep apnea, which might partially confound outcomes.

In conclusion, a few studies has explored in depth sleep architecture in MCI patients by performing polysomnographic (PSG) recordings (Brayet et al., 2016; Gorgoni et al., 2016; Hita-Yañez et al., n.d.; Maestri et al., 2015; Reda et al., 2017; Tseng et al., 2010; Westerberg et al., 2012; Yu et al., 2009), with no clear consensus as to typical robust changes, most studies either not reporting significant changes in sleep or findings that represent statistical trends.

It is unclear whether sleep is disrupted in aMCI and if present, whether sleep disruptions are associated with memory impairment. Little research has been made to determine if impaired sleep patterns appear years before AD diagnosis, so whether altered sleep patterns precede AD diagnosis remains controversial to date. None of the studies analyzed systematically examined connections between sleep macro-architecture and cognitive outcomes in detail, so specifically investigating this association is required. Further PSG studies including polygraphy respiratory and movement measures are needed to definitively confirm whether altered sleep patterns observed in MCI patients have a neurodegenerative basis or are due to sleep disorders exacerbated in healthy elders and MCI patient. In the present study we tried to address some of these limitations in previous research.

Materials and Methods

Participants

We collected data (See Table 1) from 15 aMCI patients and two control groups: 15 healthy age matched elderly subjects, and 11 AD patients[1] (see disclaimer in supplementary). Data between June 2018- Dec 2020 was collected and scored by a previous student in the lab Omer Sharon (9 MCI and 13 controls), while I collected and scored data from Jan 2021 (6 MCI, 2 controls, and 11 AD).

Participants' ages ranged between 55-85 years. Patients were recruited from the Memory and Attention Disorders Center, Neurology Department, Tel-Aviv Sourasky Medical Center (TASMC), Tel Aviv, Israel as follows: individuals with subjective cognitive complaints were referred to the clinic general practitioners, neurologists, geriatricians, and psychiatrists. Those diagnosed as suffering from aMCI were then asked if they will be willing to participate in our study. Healthy controls were recruited from the community. Healthy volunteers that were found to have low (MoCA < 26) cognitive scores (n=5) were excluded from further analysis (See Table 1-1). Subject recruitment and all other procedures were approved by the Medical Institutional Review Board (IRB) at the TASMC, and all participants provided written informed consent prior to their participation in the study.

Table 1

	Age	Gender	MMSE	MoCA	Recall	Comments	
1				Healthy Controls			
1	60	F	30	26	2		
2	62	M	29	30	5		
3	69	F	30	27	3		
4	77	F	28	17	0	Excluded	Cognitive score
5	73.5	F	29	29	5		
6	72	M	30	27	3		
7	70	M	30	28	4		
8	64	M	30	28	3	Excluded	Moderate OSA
9	73	M	30	27	2		
10	68	M	30	28	4		
11	74	M	30	26	1		
12	61	F	29	30	5		
13	68	F	29	26	2		
14	65	F	30	30	5		

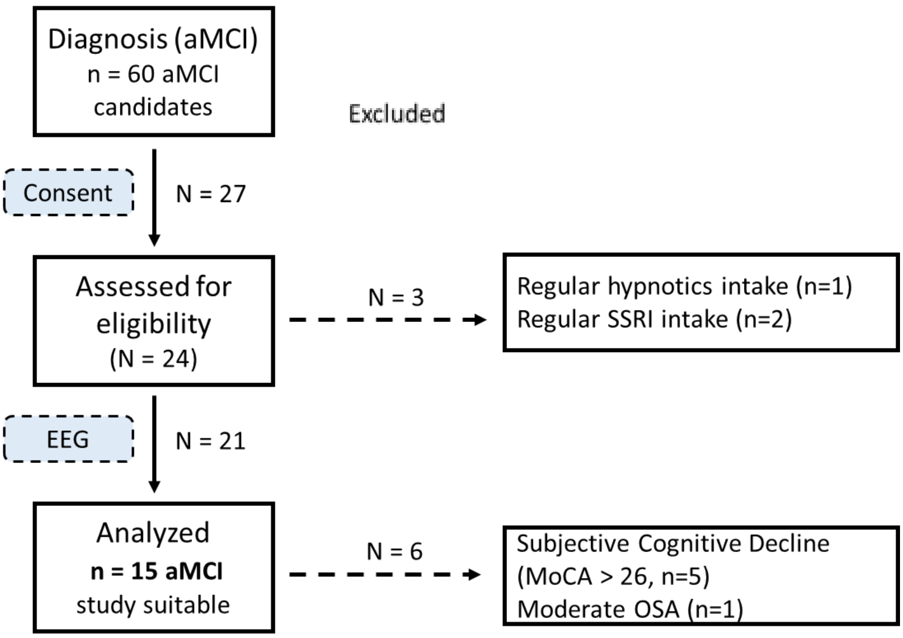
15	67	F	30	30	5		
16	60	M	27	23	1	Excluded	Cognitive score
17	69	F	30	27	5		
18	74	F	30	23	1	Excluded	Cognitive score
19	68	F	29	29	5	Excluded	Intermittent Hypoxia
20	80	M	29	22	0	Excluded	Cognitive score
21	72	M	27	23	2	Excluded	Cognitive score
22	72	M	27	27	3		
				aMCI Patients			
1	73	M	26	22	0		
2	65.5	F	30	29	4	Excluded	Cognitive score
3	71	M	29	22	2		
4	66	M	26	29	4	Excluded	Cognitive score
5	72	F	21	19	0		
6	72	M	20	16	0		
7	85	F	27	23	1		
8	64	F	26	25	3		
9	71	M	28	28	5	Excluded	Cognitive score
10	75	F	30	19	5		
11	76	F	29	27	4	Excluded	Cognitive score
12	67	M	23	17	0		
13	55	M	30	25	1		
14	64	F	29	24	0		
15	72	F	26	24	0	Excluded	Moderate OSA
16	70	M	29	19	0		
17	63	M	29	23	0		
18	67	M	26	21	0		
19	75	F	25	22	0		
20	70	F	26	26	4	Excluded	Cognitive score
21	73	F	27	20	0		
3				AD Patients[1]			
1	76	M	29	27	5		Cognitive score, AChE intake
2	72	F	24	20	0		AChE intake
3	71	F	21	17	1		AChE intake
4	84	M	27	21	1		SSRI intake
5	75	M	24	20	0		SSRI intake, AChE intake
6	63	F	9	4	0		AChE intake
7	76	M	18	13	0		
8	78	M	25	22	0		
9	85	F	28	21	0		hypnotic intake, AChE intake

10	57	M	17	13	1		AChE intake
11	61	M	23	17	1		AChE intake

***Collected Data for the aMCI Project*:** Data collected between June 2018- June 2022.

Inclusion criteria were that participants can provide informed consent and volunteer to complete an overnight sleep EEG study. Exclusion criteria included any history of sleep apnea, psychiatric disorders (e.g., major depression, bipolar disorder, schizophrenia), neurological disorders, stroke or head injury; current alcohol or drug misuse; trans-meridian travel within the last week; use of medication known to affect sleep EEG profiles (e.g., hypnotics ('sleep pills') or selective serotonin reuptake inhibitors (SSRI), lithium).

Figure 2



***aMCI Recruitment flow chart*:** 60 aMCI patients were referred to our sleep lab through the Memory and Attention Disorders Center, out of which 27 patients provided consent to go through eligibility screening, out of which 3 were excluded due to regular medication intake. 21 patients went through the entire procedure, 6 datasets were not used in the study due to cognitive or breathing issues discovered after analysis.

Procedure

Cognitive testing

Participants were invited to the lab 3 hours before their regular bedtime. Upon arrival we conducted a cognitive assessment including Mini Mental State Examination (MMSE) test and a Montreal Cognitive Assessment (MoCA, (Nasreddine et al., 2005)).

Movie viewing paradigm

After applying high density EEG (hd-EEG), participants completed a movie viewing paradigm to better reveal changes in overnight memory consolidation by using life-like video clips to engage and better address elderly and demented populations.

One hundred and twenty short movies (duration of 3.5-26 seconds) were divided into two consecutive sessions; two subsets of 100 video clips, each subset containing 80 repeating movies and 20 novel ones. Each subject watched both subsets in the same order, while the order of movies within each subset was randomized for each subject. At the first session of the paradigm (encoding phase performed in evening before sleep) a first subset of 100 movies was presented, after which patients went to sleep.

When waking up in the morning, participants completed the second viewing session of the movie paradigm (memory test performed in morning after sleep), similar to the first, only with a different subset of 100 movies. After each movie participants were asked whether they have seen the movie before.

Figure 3

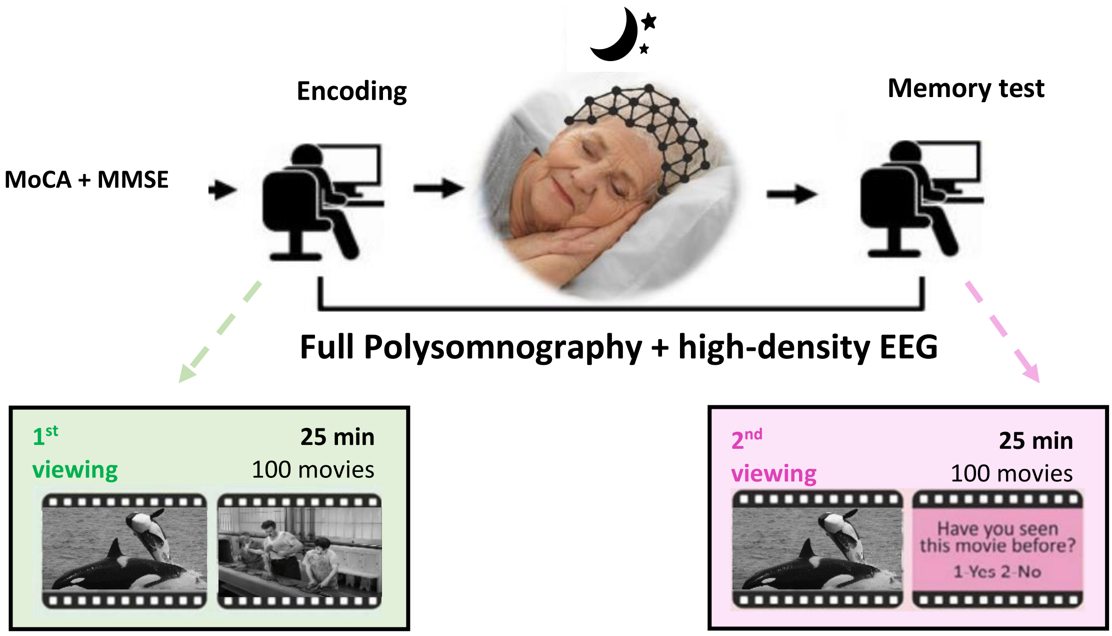


Illustration of the elderly/aMCI study design. Experimental design left to right. In the lab, the participant undergoes a cognitive assessment including Mini Mental State Examination (MMSE) and Montreal Cognitive Assessment (MoCA). Afterwards subjects complete a 1st session of the movie memory task; encoding phase. The participant then goes to sleep with full high-density EEG. After waking up in the morning participants perform 2nd session of the movie memory task; memory test phase.

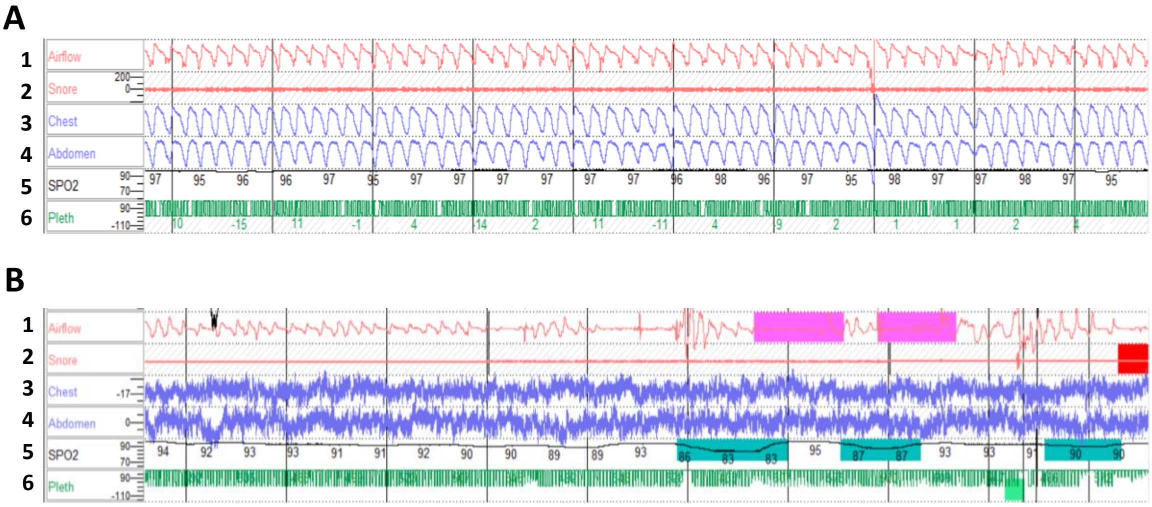
EEG Data Acquisition

We preformed High-density PSG that was recorded continuously throughout the night using a 256-channel hydrocel geodesic sensor net recording system [Electrical Geodesics, Inc. (EGI)]. Each carbon-fiber electrode, consisting of a silver chloride carbon fiber pellet, a lead wire, and a gold-plated pin, was injected with conductive gel (Electro-Cap International).

Signals were referenced to Cz, amplified via an AC-coupled high-input impedance amplifier with an antialiasing analog filter (NetAmps 300, EGI), and digitized at 1000 Hz. Electrode impedance in all sensors was verified to be <50 kΩ before starting the recording.

Breathing channels were collected including blood saturation with SpO2 sensor, nostrils airflow, respiratory effort with chest and abdomen belts. Monitoring was performed by The Sieratzki-Sagol Center for Sleep Medicine at TASMC. During breathing assessment (See fig. 4) apnea-hypopnea index (AHI) was obtained, subjects diagnosed with breathing issues (moderate to severe OSA (AHI≥15), CSA or hypoventilation, see Fig. 4-B) were ruled out.

Figure 4



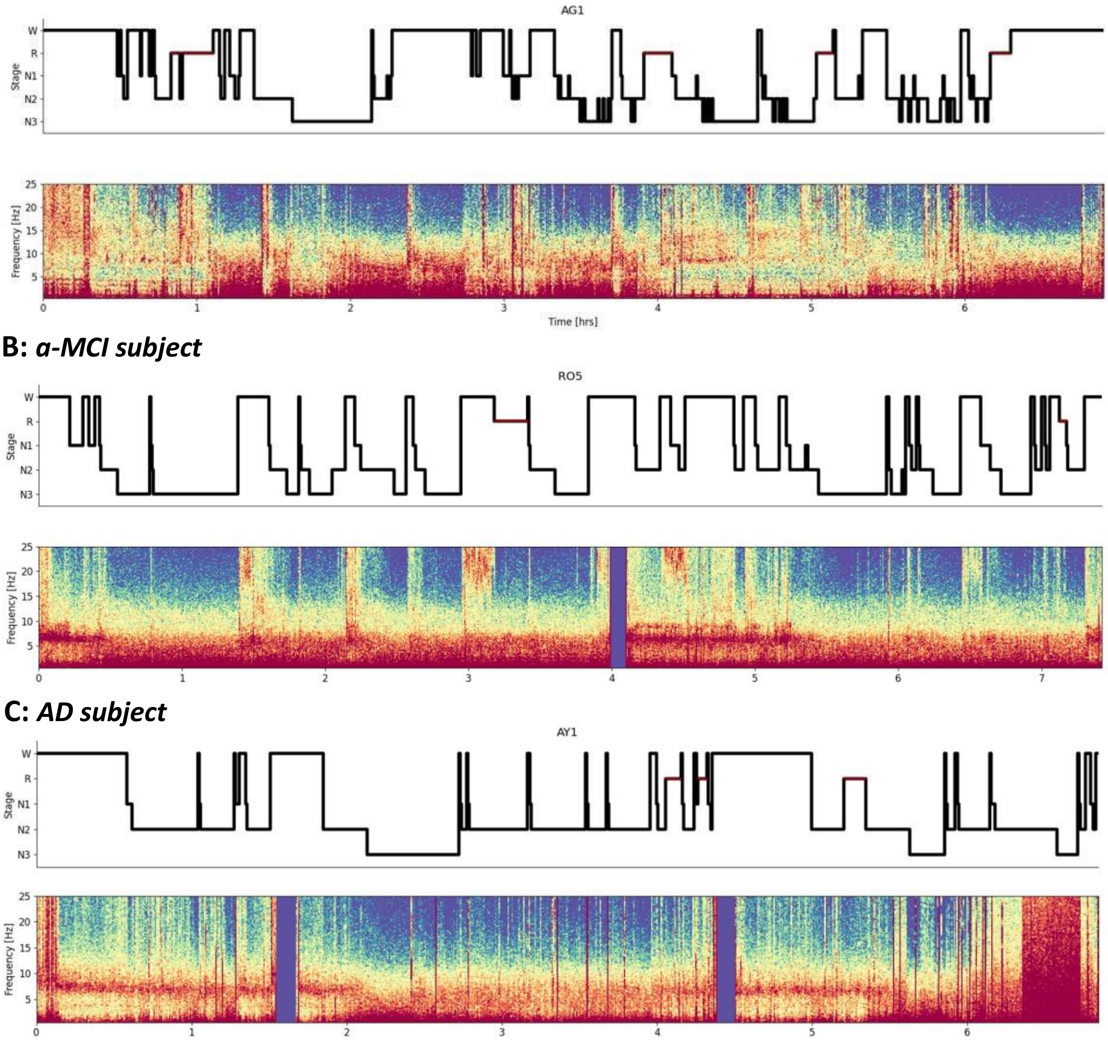
Raw breathing channels examples: Examples for breathing representation and monitoring in Alice clinical sleep scoring program including **1** Nose airflow signal (red) **2** Snore signal (red) **3** Chest breathing movement signal (blue) **4** Abdomen breathing movement signal (blue) **5** SpO2- blood oxygen levels (%;black) **6** Pleth; SpO2 intactness (green), for (**A**) Healthy subject with regularity in channels signal (**B**) Moderate OSA subject with airflow inconsistency (marked in pink) and SpO2 blood saturation drops (marked in blue).

Sleep scoring

Sleep scoring was performed according to the established guidelines of the American Academy of Sleep Medicine (Iber et al., 2007) with the aid of the sleep module in Visbrain Python package (Combrisson et al., 2017). To this end, EEG data from F3/F4, C3/C4, and O1/O2 was re-referenced to the contralateral mastoid, resampled to 250 Hz, and visualized in 30s epochs together with synchronized EOG from electrodes E1&E2 above and below eyebrows and EMG from submental electrode. Successful scoring was further verified by inspecting the time-frequency representation (spectrogram) of the Pz electrode (not involved in scoring process) superimposed with the Hypnogram.

Figure 5

A: ***Control subject***



Single Participants Recording Overview. Example hypnogram (top) of a single participant scored according to AASM guideline and spectrogram (bottom) showing entire overnight recording of a single participant from (A) Control (B) MCI (C) and AD groups.

Filters were applied including a notch filter (50 Hz), on EEG data a bandpass filter of 0.1–40 Hz and on EMG a bandpass filter of 10-100 Hz. Breathing data such as blood saturation (pleth and SpO2 bandpass filter 0-30 Hz), nostrils airflow (snore bandpass filter 10-100 Hz, airflow bandpass filter 0-15 Hz), chest and abdomen belts (belts bandpass filter 0-15 Hz) was recorded as well (See Fig. 4).

Memory performance

We quantified overnight memory consolidation by calculating the sensitivity of recognition memory index d' (Green and Swets, 1966) as a standard measure of correct recognition memory (Hits, correctly recognizing in the morning movies that were indeed presented in evening) and mistakes reporting novel movies as familiar (False Alarms, erroneously tagging lures shown in morning as movies that appeared in evening), as is often performed in studies using still images e.g. (Squire and Wixted, 2011).

Statistical Analysis

Statistical tests were conducted using ANOVA contrasts and corrected for multiple comparisons using the False Discovery Rate (FDR) (Benjamini and Hochberg, 1995). Correlations were calculated using the non-linear Spearman correlation coefficient (Spearman, 1904). Data is expressed as M=Mean, SD=Standard Deviation

Results

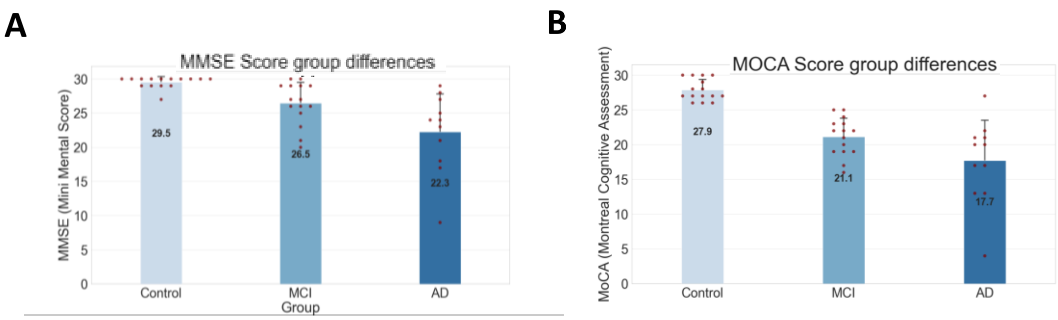
We collected data (see Table 1) from 15 Amnestic Mild Cognitive Impaired (aMCI) patients (53.3% Females, mean age=69.7), 15 healthy age matched controls (46.7% Females, mean age=68.2) and 11 ADD patients[1] (36.4% Females, mean age=72.55, Table 1). The groups did not differ significantly in aspects such as age or gender (non-significant, see Table 8-1*Demographics*), nor breathing condition (Table 8-4*Breathing*); AHI did not significantly correlate with any cognitive or memory scores, nor with any of the sleep parameters.

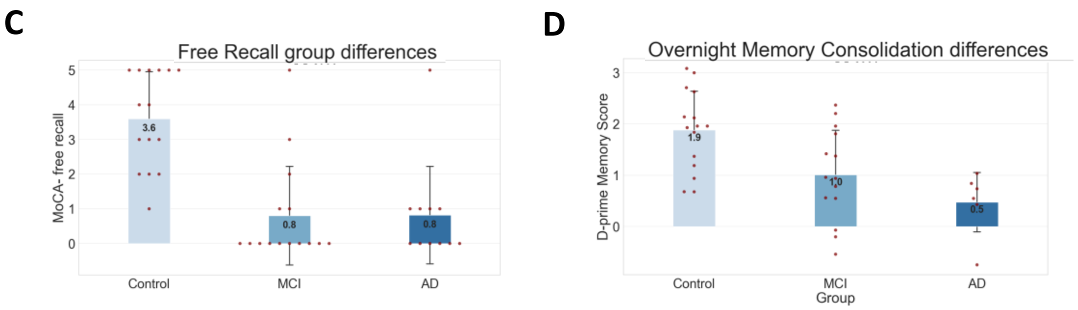
Memory performance

Healthy elderly age-matched controls had an average Mini Mental State Examination score of 29.5 (SD=0.8) while aMCI patients had an average score of 26.5 (SD=3.1) and AD subjects had a MMSE average of 22.3 (SD=5.8, p=<0.001, see Table 8 for descriptive statistics, Fig. 6 for visuals). In the MOCA test, healthy elderly had an average score of 27.9 (SD=1.6) while the aMCI had a lower average MOCA score of 21.1 (SD =2.8) and AD had 17.7 (SD=6.1, p=<0.001, Table 8). Focusing on the delayed recall section from the memory questions in the MOCA, elderly controls recalled 3.6/5 (SD=1.4) words correctly while MCI patients recalled 0.8/5 words on average (SD=1.5) and similarly AD subjects delayed recall scores were 0.8/5 (SD=1.5, p=<0.001, Table 8-2*Cognition*).

A similar profile of degraded memory performance in aMCI and more so in AD was also evident in the overnight memory consolidation using movie stimuli. Accordingly, healthy elderly had a d' score 1.9 (SD=0.8) while aMCI patients had significantly lower memory specificity, with a d' score of 1.0 (SD=0.9) and AD subjects had an average score of 0.5 (SD=0.6, p=0.005, see Table 8).

Figure 6

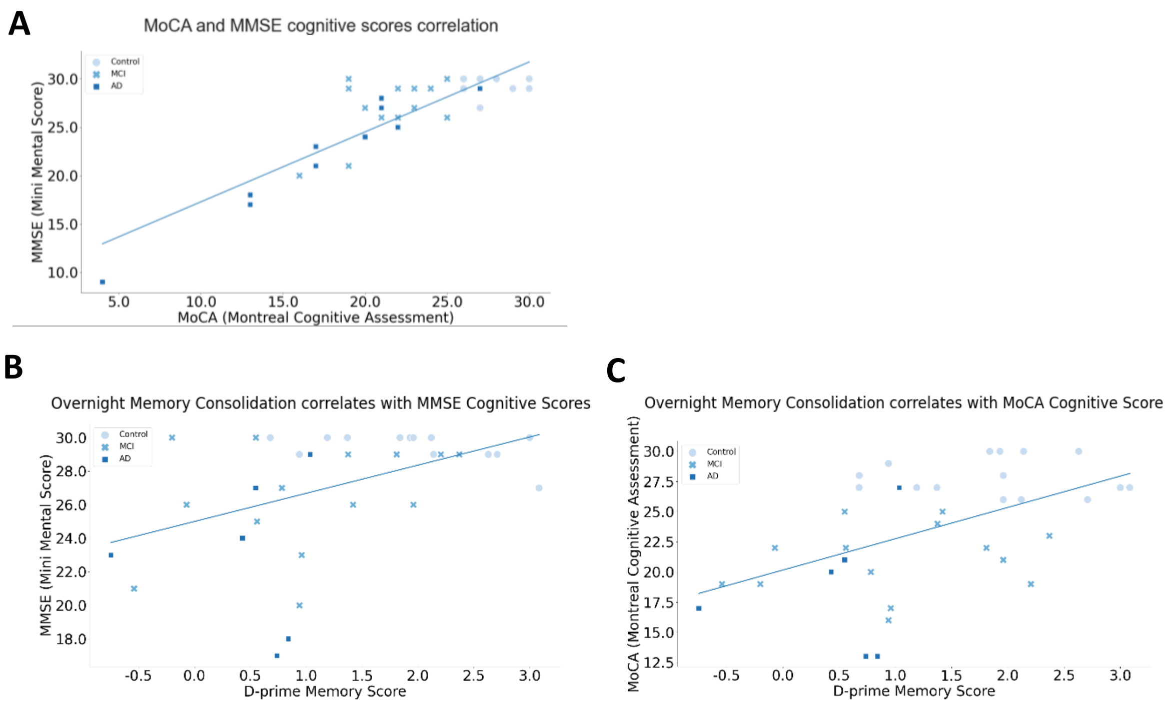




memory and cognitive scores: Comparison of averages and data distribution of clinical parameters (**A**) MMSE (**B**) MoCA (**C**) MoCA-free recall (**D**) D-prime between the groups.

MMSE scores were correlated with MOCA (r =0.78, $p < 0.001$, Fig. 7-A). In addition, overnight memory consolidation in the movie viewing paradigm correlated with both MMSE scores (r = 0.35, p = 0.042, Fig. 7-B) and with MOCA scores (r =0.49, p =0.0025, Fig. 7-C).

Figure 7



Memory and cognitive scores correlations: Correlation raw data points divided by groups and the corresponding trend line for (**A**) MMSE and MoCA cognitive scores (**B**) d-prime and MMSE scores (**C**) d-prime and MoCA scores.

Together, memory performance as assessed by standard questionnaires, as well as by our own overnight memory consolidation paradigm, confirmed degraded memory in aMCI, and more so in AD, compared with healthy controls.

Sleep architecture

When comparing sleep architecture between the groups (see Table 8 for descriptive statistics) we found that the only difference that significantly distinguished between aMCI patients and controls is REM onset latency (aMCI: M= 173 min, Control: M=105 min, p=0.008, see Table 8-3*Sleep*, fig. 9 for visuals). By contrast, AD patients presented a wide array of significant degradations in their sleep architecture, differing from age matched controls by reduced total sleep time TST (AD: M=252 min, Control: M=358 min, p=0.003), reduced sleep efficiency SE (AD: M=60%, Control: M=79%, p=0.004), N3 (AD: M=46 min, Control: M= 104 min, p=0.0004), REM (AD: M=24 min, Control: M=56 min, p=0.01) and REM onset latency (AD: M=191 min, Control: M=105 min, p=0.007). AD patients showed significant difference from aMCI patients solely regarding time spent in N3 (AD: M=104 min, aMCI: M=86 min, p=0.016).

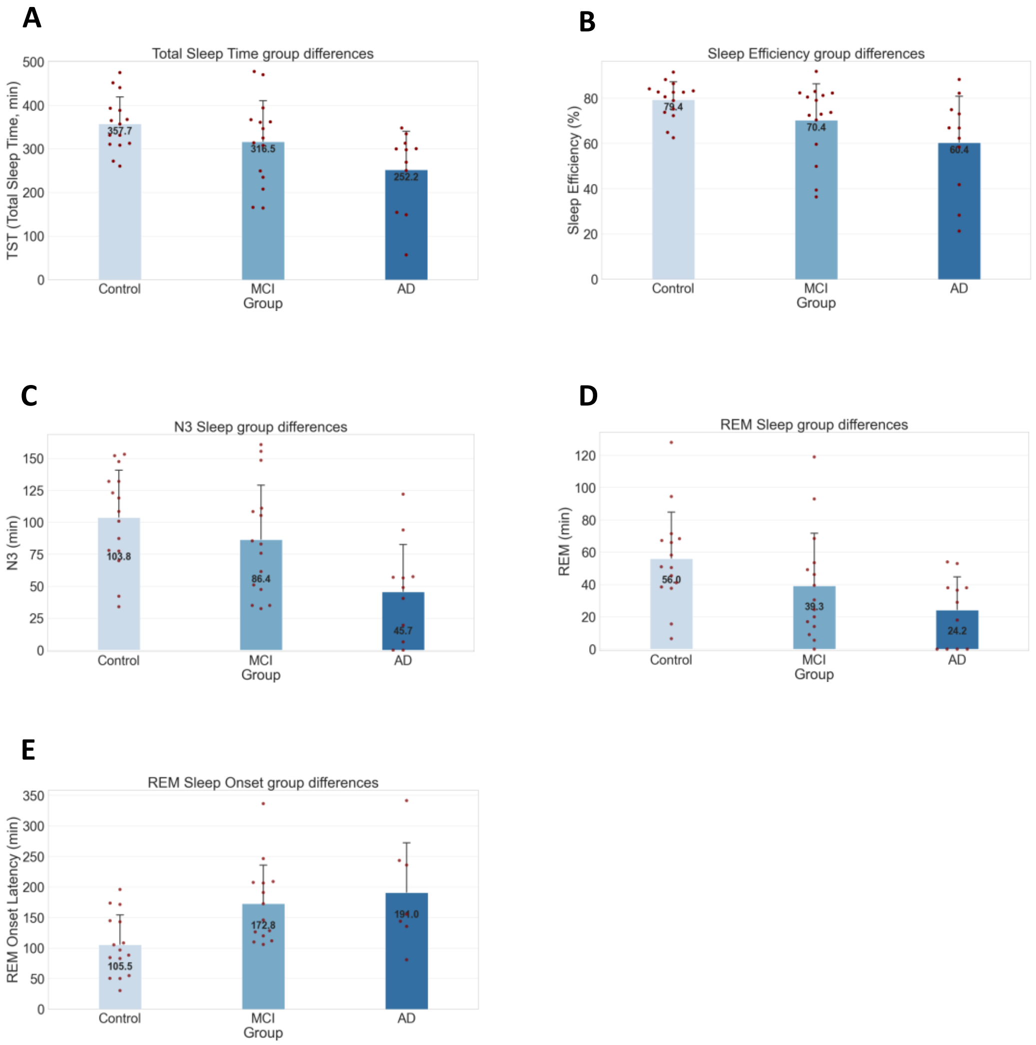
Table 8

	Group					
	Controls *(n=15)* *Mean(SD)*	**MCI patients** *(n=15)* *Mean(SD)*	**AD patients** *(n=11)* *Mean(SD)*	***P-value***		
Characteristic				*Control vs. MCI*	*Control vs. AD*	*MCI vs. AD*
1-Demographics						
Age (y)	68.23 (4.53)	69.73 (6.89)	72.55 (9.03)	0.373	0.084	0.188
Female sex (n; %)	8 (53.3%)	7 (46.7%)	4 (36.4%)	0.715	0.391	0.599
Right handedness (%)	14 (93.3%)	12 (80.0%)	11 (100%)	0.283	0.382	0.115
2-Cognition						
MMSE$^{0-30; a}$	**29.53** (0.83)	**26.47** (3.14)	**22.27** (5.82)	**0.001***	**0.001***	0.024
MoCA$^{0-30; b}$	**27.87** (1.55)	**21.13** (2.75)	**17.73** (6.08)	**<0.001***	**<0.001***	0.054
MoCA- free recall Score$^{0-5}$	**3.60** (1.40)	**0.80** (1.47)	**0.82** (1.47)	**<0.001***	**<0.001***	0.488
Overnight Memory $^{D\text{-}prime}$	**1.88** (0.78)	**1.01** (0.90)	**0.48** (0.64)	**0.005***	**<0.001***	0.104
3-Sleep architecture						
TST^{d} (min)	**357.68** (63.57)	316.53 (97.05)	**252.23** (92.18)	0.192	**0.003***	0.064
SE^{e} (%)	**79.38** (8.19)	70.39 (16.63)	**60.44** (21.5)	0.126	**0.004***	0.120
N1 (%)	11.46 (5.07)	12.06 (4.74)	20.82 (19.41)	0.881	0.035	0.048
N1 (min)	37.5 (15.13)	40.18 (20.71)	39.86 (22.56)	0.708	0.761	0.968
N2 (%)	44.28 (6.47)	46.06 (12.53)	52.83 (14.97)	0.675	0.069	0.146
N2 (min)	156.57 (29.47)	150.63 (46.46)	142.45 (66.84)	0.669	0.462	0.669
N3 (%)	28.64 (10.21)	27.56 (11.89)	18.40 (17.84)	0.824	0.058	0.089
N3 (min)	**103.81** (38.12)	**86.43** (44.17)	**45.68** (38.7)	0.248	**<0.001***	**0.016***
REM (%)	**15.20** (6.03)	11.07 (7.08)	**7.94** (7.05)	0.100	**0.010***	0.247
REM (min)	**56.02** (29.77)	39.29 (33.94)	**24.23** (21.57)	0.128	**0.010***	0.205
SO^{g} (min)	20.70 (10.42)	20.00 (24.37)	60.44 (21.5)	0.930	0.069	0.058
REM SO (min)	**105.48** (50.52)	**172.76** (65.59)	**191.00** (87.64)	**0.008***	**0.007***	0.546
4-Breathing						
AHI^{h}	5.10^{i} (3.00)	6.39^{j} (4.01)		0.164^{k}		

Note: *****: Withstood Benjamini-Hochberg FDR correction.
MMSE (Mini Mental Score)^{a} MoCA (Montreal Cognitive Assessment)^{b} TIB (Time In Bed)^{c} TST (Total Sleep Time)^{d} SE (Sleep Efficiency)^{e} WASO (Wake After Sleep Onset)^{f} SO (Sleep Onset)^{g} AHI^{h} (Apnea Hypo-apnea Index) n=7^{i} n=11^{j} DF=18-2^{k}

Sleep architecture in AD, aMCI and healthy elderly: Demographic, Clinical and sleep characteristics' average values and significance results for control, aMCI and AD groups.

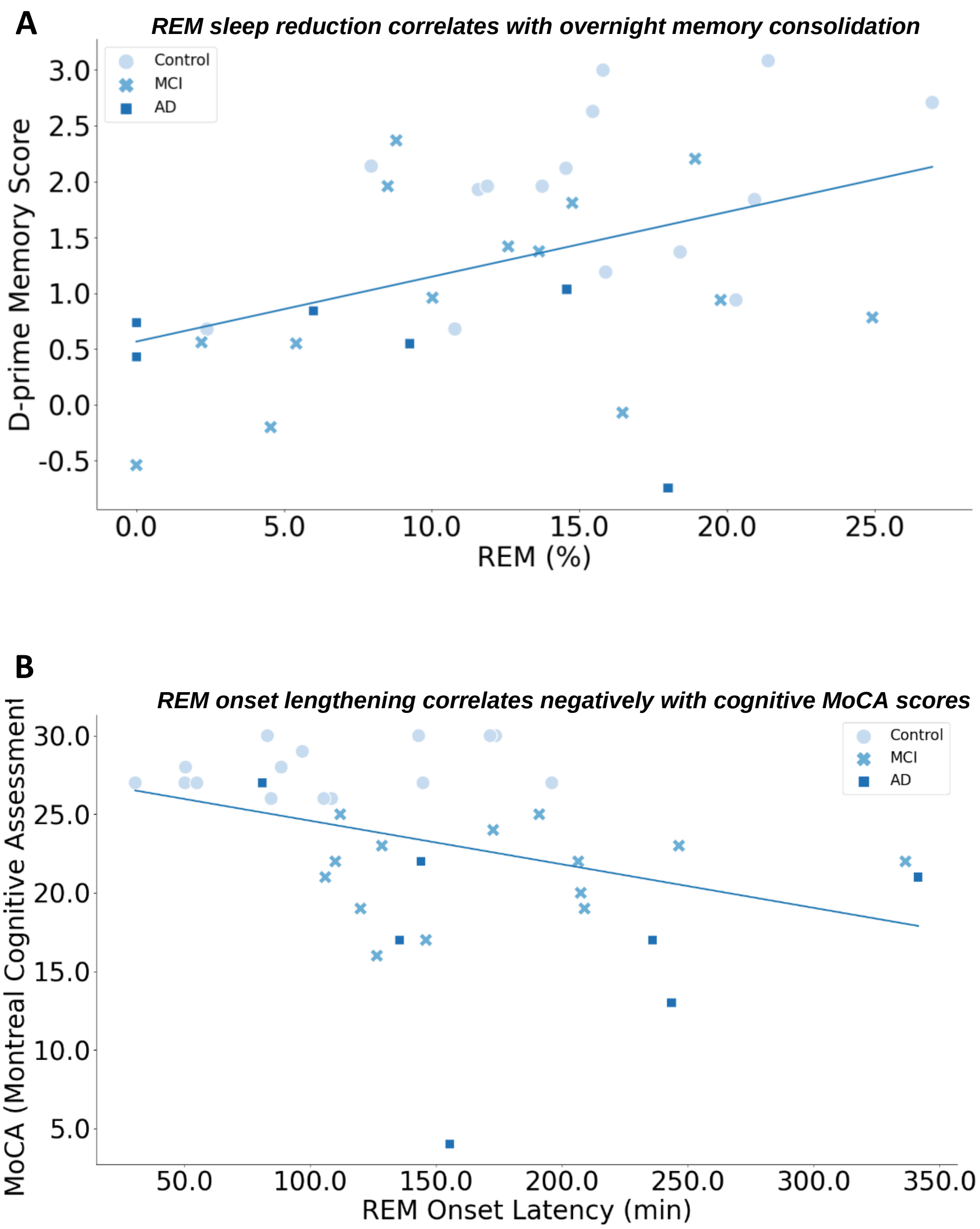
Figure 9



Sleep architecture in AD, aMCI patients and healthy elderly: Visual comparison of averages and data distribution of sleep architecture parameters (A) Total Sleep Time (min) (B) Sleep Efficiency (%) (C) N3 (min) (D) REM (min) (E) REM Sleep Onset (min).

Next, we sought to go beyond dichotomies between groups and examine potential links between individual memory performance and individual features of sleep architecture. We found a correlation between time spent in REM sleep and memory performance ($r = 0.42$, $p=0.012$, see Fig. S2-A for correlation with absolute REM time (min) as well; $r =0.34$, $p =0.045$). This correlation remained significant in a regression model as well, with age, gender and TST or N3 as covariates (Fig. S1A/B). A negative correlation was found between REM onset latency and MoCA cognitive scores ($r = -0.36$, $p=0.030$). This correlation remained significant in a regression model as well, with age and gender as covariates (see Fig. S1-C). REM sleep onset latency correlated negatively with REM amounts (r = -0.36, p =0.030, Fig. S2-B) and percentages (r = -0.43, p =0.010, Fig. S2-C) as well.

Figure 10



REM Sleep characteristics correlates with memory and cognition scores: Correlation raw data points divided by groups and the corresponding trend line (A) REM sleep reduction correlates with overnight memory consolidation (B) REM onset lengthening correlates negatively with cognitive MoCA scores.

Discussion

We found that sleep disturbances precede the clinical onset of Alzheimer disease (AD) and in early disease stages are predominantly observed in aspects related to REM sleep. Accordingly, our results highlighted longer latency to REM initiation as an exclusive aspect that significantly differed between early cognitive impairment from healthy aging. By contrast, AD dementia displayed a wide array of sleep impairments compared to age matched control group as reported in previous literature, and their differentiation from aMCI patients was expressed in the total amount spent in NREM N3 sleep.

Our results extend and refine previous studies regarding relations between sleep in aging and early dementia. Specifically, some studies (Westerberg et al., 2012) speculated that SWS begins to decrease in healthy aging and then declines further in aMCI, whereas REM sleep abnormalities are more pronounced in advanced stages of AD. By contrast, our results suggest that REM sleep abnormalities constitute the first indications in aMCI, suggesting that these are different qualitatively from both normal aging and AD dementia and characterized by REM early damage. Thus, changes in REM sleep may provide a sign of impending cognitive decline beyond regular ageing.

Our results are in line with other studies employing quantitative EEG (qEEG) in which REM sleep was pinpointed to as the preferable diagnostic marker (Brayet et al., 2016; Hassainia et al., 1997), while the regions exhibiting slower EEG frequencies in REM sleep (mostly temporal) match areas associated with early AD-related pathologies (Durand-Martel et al., 2010). Moreover, REM sleep's qEEG in mild AD subjects was correlated with their cognitive decline (Montplaisir et al., 1996), a result that was not implicated in wakefulness qEEG in those subjects.

Tseng et al.'s group found similarly, although not significantly, an increase in REM latency and decrease in the number of REM episodes (Tseng et al., 2010). What more, it was found that as the severity of cognitive impairment increased, there was a trend toward increasing REM latency and decreasing total number of REM.

Less REM sleep has been linked with deteriorating cognitive functioning in elders in general, based on population-based cross-sectional and longitudinal studies (for the Osteoporotic Fractures in Men (MrOS) Study Group, 2015). Furthermore, each 1 percentage decline of REM sleep was associated with a 9% elevated risk for incident dementia (Pase et al., 2017).

REM latency increase in AD patients was already published in the ‘90 as a prominent biomarker of AD sleep. Bliwise et al. at 1989 have not found REM percentage differences between AD and control but did notice a robust REM latency increase. And yet, in (Moe et al., 1995) paper about sleep patterns in AD, REM deficits in AD’s sleep impairment were not mentioned. Furthermore, the unique association found between REM latency to cognition that appeared in their data is implied to drive from a different correlator found- increased wakefulness, although it was less significant.

Our results involving MCI somewhat contradict Moe’s conclusion; differences in wakefulness after sleep onset differences did not emerge in our MCI patients, and correlation between WASO and REM latency, and with cognitive domains are non-significant ($r<0.25$). Interestingly, Moe et al.’s study dealing with various AD stages concluded that REM percentage is the best predictor for dementia level, emphasizing even further a possible causal rule starting with an impairment in REM latency, deriving in later stages less REM sleep and a worsening and progression of the disease.

It seems that although REM issues in AD sleep are known for years, most attention was drawn to NREM deficits, perhaps due to cholinergic drugs consumption by most of AD patients, shadowing possible conclusions regarding REM sleep. Spotting prominent REM changes patterns in early-stage drug-naïve patients positions REM sleep impairments in a new light, spotlighting them as a matter of great interest in AD’s early pathology and as a unique marker.

General sleep anatomy data framework

At 3 years prior to a diagnosis of AD atrophy is not observed outside the temporal lobes, with a sparing of the parietal or frontal regions (Whitwell et al., 2007b). A few previous studies have similarly failed to spot any atrophy of the frontal lobes in aMCI subjects (Pennanen et al., 2005), which suits their much reserved functioning. This corresponds with the Bra’ak stages, that describe neurofibrillary tangles first appearing in the entorhinal cortex and the hippocampus (transentorhinal stages I–II), spreading out into the amygdala and basolateral temporal lobe (limbic stages III–IV); stages that seem to correspond with typical damage described in aMCI, and finally continuing their progression into the isocortical association areas (isocortical stages V–VI), in which a pathological diagnosis of high-probability AD is given.

At the time point of a clinical diagnosis of AD the pattern of cerebral loss is dramatically widespread with severe status of the medial temporal lobes and the temporo-parietal association cortices and a beginning of substantial involvement of the frontal lobes, all typically involved in AD (Matsuda et al., 2002). These widespread patterns of atrophy seem to parallel the diminishing cognitive functioning of AD. These findings logically suit our results regarding sleep, reflecting progression from MCI to AD as a process of restricted damage in certain domains spreading over time; worsening the existing symptoms in REM sleep and progressing into new ones in other sleep regions and stages and diminishing general sleep quality.

There is a continuous range between “normal” aging and AD dementia, with the amyloid plaque build-up tending to form mostly before cognitive deficits onset, at a transitional stage from healthy aging, while neurofibrillary tangles and neuron loss occur in parallel with the progression of cognitive decline (Serrano-Pozo et al., 2011). In the initial steps when amyloid gather but still do not result in pathological damage to neurons, it is unclear how and whether it affects cognition. Opposed to that, Sleep diminishes as well before cognitive issues onset in parallel with the toxin buildup and sleep is correlated strongly with cognition. Perhaps in those initial steps amyloid neurotoxicity affects sleep, which in turn could be the direct reason causing cognitive deterioration at early stages. What could be the possible mechanisms that may support such a hypothesis?

Cholinergic system

The initiation of REM sleep depends on the activation of cholinergic neurons in the brainstem; lateral dorsal tegmental nuclei (LDT) and pedunculo-pontine tegmental nuclei (PPT) (Yu et al., 2009). The pontine cholinergic network is an important REM generator mechanism (Vanni-Mercier et al., 1989), and REM sleep disturbance could result from damage to pedunculopontine tegmentum (PPT) cholinergic neurons, data supported by AD animal models (Zhang et al., 2005). Morphological changes were observed in mesopontine cholinergic neurons of the pedunculopontine and laterodorsal tegmentum (Mufson et al., 1988), manifesting a strong association to REM deficits in AD patients (Hita-Yañez et al., n.d.).

The widespread disruption of the cholinergic system, which is active during REM sleep, already exists in minor form in MCI. Therefore, cholinergic system impairment (McGeer et al., 1983) may lay in base of abnormal REM sleep structure in MCI (Marcone et al., 2012), and is correlated with dementia severity (Baskin et al., 1999; Coyle et al. 1983).

The nucleus basalis of Meynert (NbM; Hanyu et al., 2002) in the Basal Forebrain provides the most significant amount of acetylcholine to the neocortex during REM sleep (Whitehouse et al., 1981) playing a major part in cortical EEG activation characteristics of REM state. Converging evidence shows that cholinergic neurons of nucleus basalis of Meynert are selectively vulnerable to neurodegeneration in AD patients (Whitehouse et al., 1981).

This hypothesis could be extended to preclinical stages of AD, MCI patients who exhibit volume reductions of the NbM, a parameter that was found to correlate with impaired cognition in this population (Grothe et al., 2010) and strengthen hypothesis about an early vulnerability to neurodegeneration in cholinergic pathways, which are known to be involved in REM sleep physiology (Grothe et al., 2010). Therefore, it seems sensible that cholinergic deficits originated in the NbM were partially linked to REM deficits spotted in MCI and exacerbated in AD patients.

In line with mentioned above, cholinesterase inhibitors, that are the primary drug choice in dementia and MCI treatment, increase REM sleep quality and duration. Most interestingly, the level of memory improvement in treated AD patients can be predicted by the degree of achieving improvement in REM sleep parameters as result from the drug (dos Santos Moraes et al., 2006). Unfortunately, as to now, there are limited options for selectively increasing REM sleep.

NO

Inducible nitric oxide synthase (iNOS) produces nitric oxide has an intriguing connection to our results. Increasing evidence indicates that nitric oxide (NO) plays an important, yet complex, role in the regulation of rapid eye movement (REM) sleep

(Zhang et al., 2005), iNOS appears to be vital for both initiation and maintenance of REM in ageing animals (Colas et al., 2005). Patients with AD increase iNOS expression significantly (Tseng et al., 2010); and this irregularity in the NO system might alter sleep architecture and add to behavioral disturbances (Tseng et al., 2010).

In Tseng et al. study, iNOS mRNA significantly increased during REM sleep of AD patients. Elevated iNOS amounts in REM sleep in AD patients may indicate a compensation mechanism for maintaining REM sleep. Higher expression of iNOS was also associated with worse cognitive impairment in Alzheimer's Disease (AD) patients compared to controls (Fernández-Vizarra et al., 2004).

Importantly, iNOS over-expression is a known sign of current inflammation, which is thought to be a factor in the emergence of AD, and clearing iNOS was demonstrated to reverse amyloid β–related cognitive decline (Medeiros et al., 2007). Tseng et al. data suggests that the heightened iNOS during REM sleep in AD patients could have a two-fold impact; While The attempt for REM maintenance that is bolstered by iNOS could help support REM dependent memory consolidation, the iNOS-caused inflammatory processes may result in further neurodegenerative damage, leading to AD.

This describes how REM sleep is maintained through iNOS mechanisms, and how impaired REM sleep could be causally involved in early steps of the development of AD pathogenesis, however further research is needed to fully understand its role.

Possible mechanisms underlying early REM impairment

There are few possible theoretical mechanisms explaining possible connections between REM sleep deficits in early stages, expressed in initial steps as a constrained memory impairment, as an explanatory causal factor for the evolvement of neurodegeneration, AD and widespread damage and broad functional impairments later.

A possible explanation for the absence of NREM alterations in our results in MCI patients is that the brain damage responsible for those changes emerges only in diagnosed AD, which is characterized by neuropathological alterations to cortical areas and subcortical regions. Age-related medial prefrontal cortex (mPFC) gray-matter atrophy for example is associated with reduced NREM SWA in older adults (Mander et al., 2013), strengthening the theory claiming early pathology is focused on REM related sensitive areas, while NREM deficits evolve later, representing a widespread general diminish of grey matter tissues

However, it is important to note a weakness of our study, deriving from usage of AASM's traditional scoring system, which is not ideally suited for certain populations, such as elderly, that prominently experience sleep diminishment, characterized by a weakened intensity of various electrical patterns in sleep. Among those is SWS, for which AASM's criteria places a strict requirement as for slow waves' amplitude (>80 μV), regardless of age. This complicates the possibility to detect altered SWS in this population, and any changes in density and amplitude in the weaker slow waves that define the elders.

Implications

Sleep disruption occurs in most cases before the eruption of relevant neuropsychological impairments (Maestri et al., 2015) and so over the years, great efforts have been made to find new markers in order to diagnose AD at an early stage, predict its outcome and possibly prevent or revert the chain of events leading to AD neurodegeneration, with little success.

Depression and other psychiatric disorders are associated with REM sleep disturbance (Spoormaker and Montgomery, 2008), and the typical finding is that REM latency is decreased in pseudodementia, which is thought to result from cholinergic suprasensitivity, particularly in depressive diseases (Gillin et al. 1982; Dube).

In the neurological field, sleep disturbances are frequently the first sign of neurological disorders that interfere with sleep in general, but less is known specifically about REM latency in dementia. Changes in REM sleep different domains are well known of in certain neurological diseases, many years before clinical onset, such as RBD in Parkinson and Lewy body disease, related specifically to movement regulation in REM stage.

All those emerging evidence points to REM sleep stage in general as a sensitive possible early neuropsychology pathology marker, presenting disruptions in REM sleep that precede disease eruption in many years. Since REM sleep modulation depends on cholinergic systems (Gillin and Sitaram 1984) and Alzheimer’s disease is a cholinergic suppressive neurodegenerative disease, makes sense REM latency should be lengthened in AD, and this provides a unique and salient marker among neurological diseases.

The lack of reliable markers for early AD detection is a major obstacle for the development of treatments that could delay the progression of AD disease. Identification of individuals at risk at the preclinical phase would drastically improve the prevention of neurological deterioration (Brayet et al., 2016). REM latency could possibly be a relevant and unique surrogate marker for pre-dementia Alzheimer disease. Although qEEG REM markers have been established (Brayet et al., 2016), here we suggest a more available REM characteristic to produce quickly and possibly automatically AD markers.

Altogether, we found that alterations in sleep patterns do occur years before the diagnosis of Alzheimer's disease, above and beyond what is seen in healthy aging and could signal neurodegenerative pathology. These findings help unfold the boundaries separating pathological from healthy aging. Future research could use those results to develop sleep markers assisting in earlier detection of this dementia and for managing sleep disturbances in populations at risk during (and wishfully, preventing) their progression to AD. REM sleep has potential to reveal new knowledge in the field of neurodegeneration and to create EEG markers that can identify the transition to AD dementia.

Future Research

Future work should quantitatively analyze the EEG, examining REM and NREM sleep to obtain the full profile of abnormalities in electrical brain activity during sleep that are beyond sleep architecture. It would be intriguing to discover if REM sleep is already displaying signs of deterioration in this aspect as well, such as altered power or timing of theta activity or invasion of slow waves (‘EEG slowing’). Or perhaps on the contrary; enhanced theta frequency by compensatory mechanisms enacted in the aMCI stage to make up for the delayed onset of REM sleep. Regarding NREM sleep, detailed analysis of power and timing of slow wave and spindle activities is due.

Additional directions for future studies could be to relate the sleep finding observed here to longitudinal monitoring of disease progression, and to investigate the relation with other disease biomarkers including structural brain imaging and CSF samples.

Supplementary

[1]Disclaimer: AD Patients were added to the project in a later stage of analysis, as a positive control group, to establish that when more extensive sleep impairments are present then they are indeed observed and detected, and to assess the uniqueness of the results received in aMCI patients. AD patients did not go through identical exclusion procedure as aMCI (the target research population), for example with respect to apnea and to drug intake. Despite this caveat, AD sleep patterns observed here are in line with those reported in the literature.

Figure S1

A

Linear Regression ▾

Model Summary - D-prime Memory Score

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	0.964
H_1	0.485	0.235	0.133	0.898

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	7.423	4	1.856	2.302	0.082
	Residual	24.186	30	0.806		
	Total	31.608	34			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized[a]	t	p
H_0	(Intercept)	1.292	0.163		7.926	< .001
H_1	(Intercept)	2.934	1.960		1.497	0.145
	REM (%)	0.064	0.030	0.470	2.134	0.041
	TST (Total Sleep Time, min)	-5.141×10^{-4}	0.002	−0.049	−0.221	0.826
	Age	−0.034	0.026	−0.217	−1.330	0.194
	Gender (M)	0.176	0.321		0.548	0.588

[a] Standardized coefficients can only be computed for continuous predictors.

B

Linear Regression

Model Summary - D-prime Memory Score

Model	R	R²	Adjusted R²	RMSE
H₀	0.000	0.000	0.000	0.964
H₁	0.483	0.234	0.132	0.899

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H₁	Regression	7.388	4	1.847	2.288	0.083
	Residual	24.220	30	0.807		
	Total	31.608	34			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized[a]	t	p
H₀	(Intercept)	1.292	0.163		7.926	< .001
H₁	(Intercept)	2.793	1.866		1.497	0.145
	REM (%)	0.060	0.022	0.437	2.709	0.011
	N3 (%)	−0.001	0.013	−0.014	−0.082	0.936
	Gender (M)	0.185	0.333		0.554	0.584
	Age	−0.034	0.026	−0.212	−1.310	0.200

[a] Standardized coefficients can only be computed for continuous predictors.

C

Linear Regression

Model Summary - MoCA (Montreal Cognitive Assessment)

Model	R	R²	Adjusted R²	RMSE
H₀	0.000	0.000	0.000	5.562
H₁	0.374	0.140	0.060	5.394

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H₁	Regression	151.714	3	50.571	1.738	0.179
	Residual	931.036	32	29.095		
	Total	1082.750	35			

Note. The intercept model is omitted, as no meaningful information can be shown.

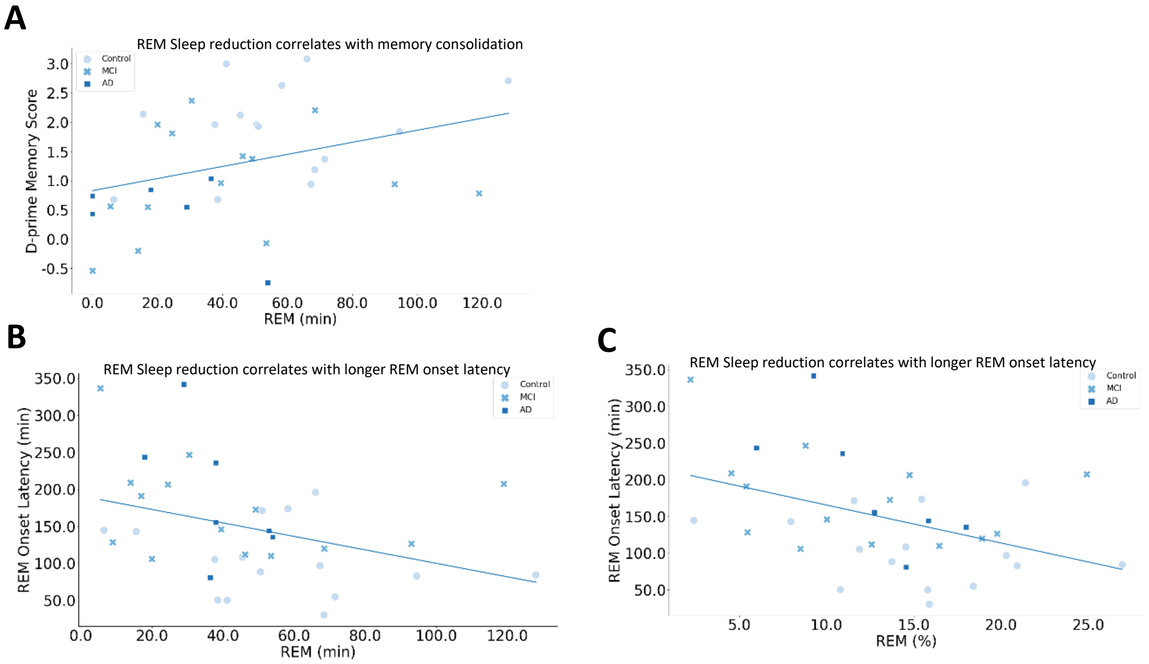
Coefficients

Model		Unstandardized	Standard Error	Standardized[a]	t	p
H₀	(Intercept)	23.250	0.927		25.081	< .001
H₁	(Intercept)	29.598	9.889		2.993	0.005
	REM Onset Latency (min)	−0.028	0.013	−0.361	−2.160	0.038
	Age	−0.025	0.145	−0.029	−0.176	0.862
	Gender (M)	−0.894	1.821		−0.491	0.627

[a] Standardized coefficients can only be computed for continuous predictors.

d-prime regression model: Regression models for d-prime and REM (%) relationship including covariates; (**A**) TST, age and gender (**B**) N3 (%), age and gender. Regression model for MoCA and REM onset latency (min) relationship including covariates; (**C**) age and gender.

Figure S2



Behavioral and sleep parameters correlations: Correlation raw data points divided by groups and the corresponding trend line for (**A**) D-prime score and REM amount in minutes (**B**) REM amount in minutes and REM onset latency (**C**) REM percentage and REM onset latency.

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